

# Micro-CFO Frontend - Product Requirements Document (PRD)

## 1. Executive Summary

**Product Name:** Micro-CFO Web Dashboard

**Target Users:** Indian MSME owners, manufacturers, proprietors, freelancers

**Primary Goal:** Provide an intuitive interface for autonomous financial compliance and growth management

**Core Experience:** Multi-channel (WhatsApp + Web Dashboard) financial assistant

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## 2. Recommended Technology Stack

**Best Tool:** Next.js 14+ with React

**Why This Stack:**

- **Next.js:** Server-side rendering for fast loading, SEO optimization, and excellent mobile performance (critical for Tier-2/3 users with slower internet)
- **React:** Component reusability across web and future mobile apps
- **Tailwind CSS:** Rapid UI development with responsive design
- **shadcn/ui:** Pre-built accessible components that look professional
- **Recharts/Chart.js:** Financial data visualization
- **Vercel:** Easy deployment with excellent performance

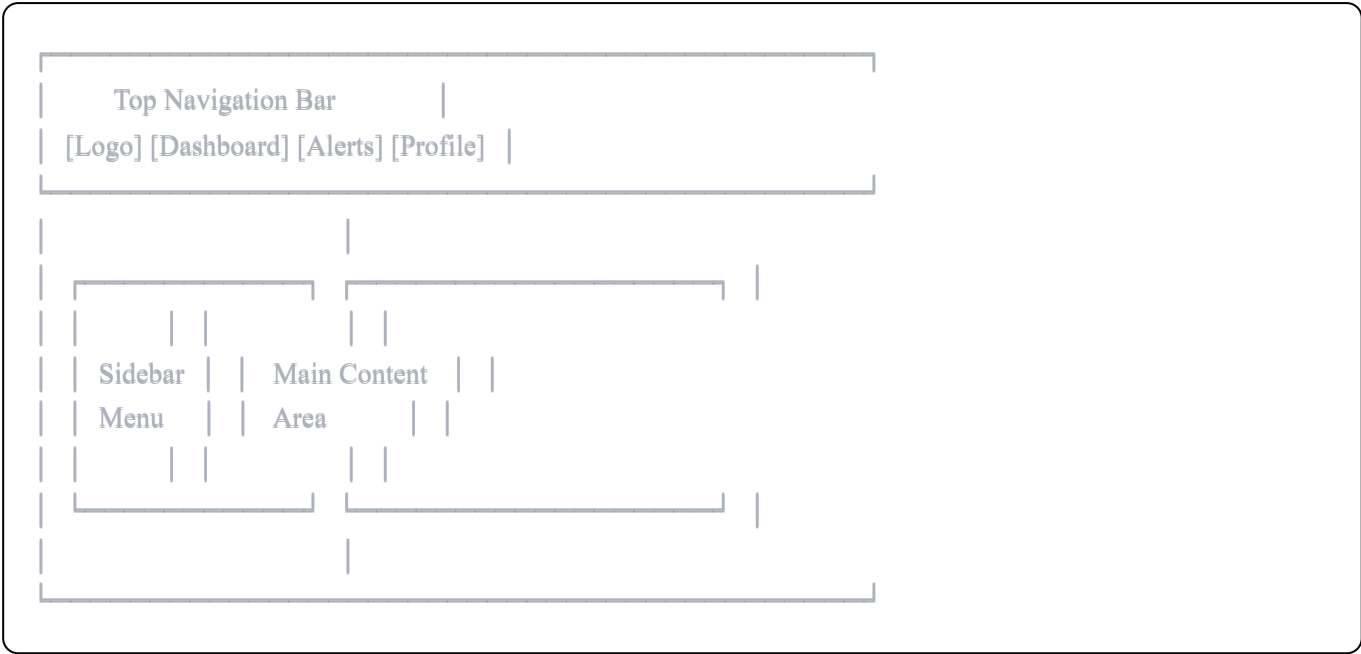
**Alternative Stack (If budget/resources are limited):**

- **Flutter Web:** Single codebase for web + mobile apps
- **Webflow + Custom Code:** For MVP if non-technical team

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### 3. UI Architecture & Components

#### 3.1 Core Application Structure



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### 4. Detailed Component Breakdown

#### 4.1 Dashboard (Home Screen)

**Purpose:** Provide at-a-glance financial health and AI agent activity

**Components:**

##### 1. Financial Health Score Card

- Visual meter (0-100 score)

- Color-coded: Red (<40), Yellow (40-70), Green (70+)
- Breakdown: Compliance %, Cash Flow Health, Subsidy Opportunities

## 2. **AI Agent Activity Panel**

- 4 Cards representing each agent:
  - **Visual Auditor:** "X documents processed today"
  - **Legislative Sentinel:** "Y alerts this week"
  - **Subsidy Hunter:** "Z opportunities found"
  - **Negotiator:** "W emails sent"
- Each card shows status indicator and last activity timestamp

## 3. **Quick Actions Widget**

- Upload Invoice button (camera icon)
- Scan Document button
- Check Subsidy Eligibility button
- Generate Compliance Report button

## 4. **Real-Time Alert Feed**

- Scrollable list of latest AI actions
- Priority color coding (Critical Red, Warning Yellow, Info Blue)
- Click to expand for details

## 5. **Cash Flow Projection Chart**

- 30/60/90 day forecast
  - Line graph with predicted inflows/outflows
  - Highlight critical dates (upcoming payments, deadlines)
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## 4.2 Document Scanner & Upload Interface

**Purpose:** Primary entry point for invoice/bill processing

### Components:

#### 1. Upload Zone

- Drag-and-drop area
- Mobile: Camera capture button
- Desktop: File browser
- Support: JPG, PNG, PDF (max 10MB)
- Preview thumbnail after upload

#### 2. Processing Status Modal

- Animated loading state ("Agent A is analyzing...")
- Progress bar (0-100%)
- Real-time analysis results as they appear

#### 3. Audit Result Card

- **Top Section:** Document preview thumbnail
- **Middle Section:**
  - Extracted data fields (Amount, Vendor, GST Number, Date)
  - Editable fields with validation
- **Bottom Section:**
  - AI Verdict:  Compliant /  Warning /  Non-Compliant
  - Detailed explanation in plain language
  - Suggested actions

#### 4. WhatsApp Integration Panel

- QR Code to connect WhatsApp number
  - Status: Connected/Disconnected
  - Option to enable/disable WhatsApp notifications
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### 4.3 Compliance Dashboard

**Purpose:** Monitor regulatory health and upcoming deadlines

**Components:**

1. **Compliance Calendar**

- Monthly view with color-coded dates
- Hover to see deadline details
- Filter by: GST, MCA, Labour Laws, Tax Filings

2. **Regulatory Updates Feed**

- Agent B (Legislative Sentinel) output
- Card-based layout:
  - Title: "New GST Amendment Affects You"
  - Summary: 2-3 sentence explanation
  - Impact Level: High/Medium/Low
  - Action Required: Yes/No (with CTA button)
  - Source: Link to official PDF

3. **Document Checklist**

- Checklist of required filings
- Status indicators:  Filed,  Pending,  Overdue

- Auto-generated reminders

#### 4. **Penalty Risk Meter**

- Visual gauge showing risk level
  - Breakdown of potential penalties by category
  - Recommended immediate actions
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#### 4.4 **Subsidy Explorer**

**Purpose:** Discover and apply for government schemes

**Components:**

##### 1. **Eligibility Matcher**

- Smart search bar: "What subsidies am I eligible for?"
- Filter options:
  - Industry sector
  - Business size
  - Location (state/district)
  - Recent transactions/purchases

##### 2. **Subsidy Cards Grid**

- Each card shows:
  - Scheme name and logo
  - Potential benefit amount (₹ range)
  - Eligibility criteria (simple bullet points)
  - Application deadline

- Match score (% based on user profile)
- CTA: "Auto-Draft Application"

### 3. Application Tracker

- Table view of submitted applications
- Columns: Scheme Name, Date Submitted, Status, Estimated Amount
- Status: Draft, Submitted, Under Review, Approved, Rejected

### 4. Subsidy Timeline

- Visual timeline of past subsidies received
- Future projections based on business activities

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## 4.5 Cash Flow & Negotiation Center

**Purpose:** Manage payments and AI-assisted communication

**Components:**

### 1. Cash Flow Forecast Graph

- Interactive chart (zoom, filter by date range)
- Predicted vs. Actual overlay
- Annotations for major events (predicted shortfalls)

### 2. Payment Collection Dashboard

- **Overdue Invoices Table:**
  - Client name, Amount, Days overdue
  - AI Negotiator status: "Email drafted", "Sent", "Responded"
- **Quick Actions:**

- "Generate Follow-up" button
- "View Email Draft" (opens modal)
- "Send Reminder" (auto-send via Agent D)

### **3. Vendor Payment Tracker**

- List of upcoming payables
- AI suggestions: "Request 15-day extension from Vendor X"
- Draft email preview with edit capability

### **4. Email Template Library**

- Pre-written templates by Agent D
  - Categories: Payment requests, Credit extensions, Invoice disputes
  - Customization options (tone: Polite, Firm, Urgent)
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## **4.6 Settings & Profile**

### **Components:**

#### **1. Business Profile Setup**

- Company name, type, GST number
- Industry sector selection
- Revenue range (for subsidy matching)
- Bank account linking via Account Aggregator

#### **2. AI Agent Configuration**

- Toggle agents on/off
- Set automation levels (Manual Review vs. Auto-Execute)



- Language preference (English, Hindi, Tamil, Telugu)

### 3. Notification Preferences

- Channel selection: WhatsApp, Email, SMS, In-app
- Frequency: Real-time, Daily digest, Weekly summary
- Priority filters: Critical only, All alerts

### 4. Data & Privacy

- Data deletion options
  - Export financial data (CSV/PDF)
  - Connected accounts management (WhatsApp, Bank, Email)
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## 5. UI/UX Design Principles

### 5.1 Mobile-First Design

- 80% of Tier-2/3 users access via mobile
- Touch-friendly buttons (min 44x44px)
- Single-column layouts for mobile
- Progressive disclosure (show advanced features on demand)

### 5.2 Vernacular Support

- Full RTL support for future Urdu/Arabic
- Font selection: Supports Devanagari, Tamil scripts
- Cultural considerations: Use ₹ symbol, dd/mm/yyyy dates, lakhs/crores format

### 5.3 Accessibility

- WCAG 2.1 AA compliance
- Screen reader support for visually impaired
- High contrast mode
- Keyboard navigation

## 5.4 Color System

- **Primary:** Trust Blue ( #1E40AF) - for CTAs, headers
- **Success:** Green ( #16A34A) - compliant, approved
- **Warning:** Amber ( #F59E0B) - review needed
- **Danger:** Red ( #DC2626) - non-compliant, critical
- **Neutral:** Gray scale for backgrounds, text

## 5.5 Typography

- **Headings:** Inter or Poppins (clear, modern)
  - **Body:** System fonts for performance
  - **Numbers:** Tabular figures for financial data alignment
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## 6. Key User Flows

### Flow 1: Upload Invoice → Get Audit Result

1. User clicks "Upload Invoice" from dashboard
2. Takes photo or uploads file
3. Processing modal appears (3-5 seconds)

4. Result card displays with compliance verdict
5. User approves → Data saved to ledger
6. Optional: Share to accountant via email

**Flow 2: Discover Subsidy → Apply**

1. User receives notification: "New subsidy match!"
2. Opens Subsidy Explorer
3. Reviews matched scheme card (95% match score)
4. Clicks "Auto-Draft Application"
5. Agent C generates pre-filled form
6. User reviews, edits if needed
7. Clicks "Submit" → Tracks in Application Tracker

**Flow 3: Cash Crunch → AI Negotiation**

1. Dashboard shows "Cash shortage predicted in 10 days"
  2. User clicks alert → Redirects to Negotiation Center
  3. Sees suggested action: "Request payment from Client X"
  4. Previews AI-drafted email
  5. Edits tone/content if needed
  6. Clicks "Send" → Agent D sends email
  7. Tracks response in dashboard
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## 7. Technical Requirements

### 7.1 Performance

- First Contentful Paint: < 1.5s
- Time to Interactive: < 3s
- Mobile optimization mandatory
- Offline mode for viewing cached data

### 7.2 Security

- End-to-end encryption for financial data
- OAuth 2.0 for Account Aggregator integration
- WhatsApp Business API compliance
- GDPR-ready (for future expansion)

### 7.3 API Integrations

- **Backend AI Agents:** RESTful API or GraphQL
- **WhatsApp Business API:** For bot functionality
- **Account Aggregator API:** For bank data
- **Government Portals:** Web scraping or official APIs (if available)
- **Email Service:** SendGrid/AWS SES for Agent D

### 7.4 Data Visualization Libraries

- **Charts:** Recharts (React-friendly) or Chart.js
- **Tables:** TanStack Table (formerly React Table)

- **Calendars:** FullCalendar or custom component
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## 8. MVP Feature Prioritization

### Phase 1 (Launch MVP - 8 weeks)

- ✓ Dashboard with health score
- ✓ Document upload + Agent A audit
- ✓ Basic compliance calendar
- ✓ WhatsApp integration (receive alerts)
- ✓ Top 5 subsidy matches

### Phase 2 (3 months post-launch)

- ✓ Full Subsidy Explorer with auto-draft
- ✓ Cash flow forecast + Agent D negotiation
- ✓ Legislative updates feed (Agent B)
- ✓ Mobile app (React Native reuse)

### Phase 3 (6 months post-launch)

- ✓ Vernacular language support (Hindi, Tamil)
  - ✓ Advanced analytics dashboard
  - ✓ Integration with accounting software (Tally, Zoho)
  - ✓ Multi-user access (for CA/team members)
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## 9. Success Metrics

### User Engagement:

- Daily Active Users (DAU)

- Average session duration
- Documents uploaded per user per month

#### **AI Effectiveness:**

- Audit accuracy rate (Agent A)
- Subsidy match conversion rate
- Email response rate (Agent D success)

#### **Business Impact:**

- Penalties avoided (₹ amount)
  - Subsidies claimed (₹ amount)
  - User retention rate (90-day)
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## **10. Design Mockup Structure**

### **Recommended Mockup Flow:**

1. **Landing Page** (Marketing site - separate from dashboard)
  - Hero section with problem/solution
  - How it works (4 agents explained simply)
  - Pricing/signup CTA
2. **Login/Signup Screen**
  - Email/Phone OTP
  - Google Sign-in option
  - WhatsApp number linking during onboarding

### 3. **Onboarding Wizard** (5 steps)

- Business details
- Connect bank (Account Aggregator)
- Upload first invoice (tutorial)
- Set notification preferences
- Dashboard tour

### 4. **Main Dashboard** (as described in Section 4.1)

### 5. **All other screens** (Sections 4.2-4.6)

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## 11. Prototyping & Development Timeline

**Weeks 1-2:** Figma mockups + user testing

**Weeks 3-4:** Setup Next.js project + component library

**Weeks 5-6:** Build core dashboard + document upload

**Weeks 7-8:** Integrate backend APIs + WhatsApp

**Week 9:** Testing, bug fixes, polish

**Week 10:** Soft launch with 50 beta users

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## 12. Why Next.js is the Best Choice

**For Micro-CFO specifically:**

1. **Performance:** Server-side rendering critical for slow mobile networks in Tier-2/3 cities
2. **SEO:** If you add content marketing, Next.js is best for Google ranking
3. **API Routes:** Can handle some backend logic (webhooks, lightweight processing) without separate server

4. **Image Optimization:** Automatic optimization for invoice/document images
5. **Easy Deployment:** Vercel makes deployment trivial
6. **Ecosystem:** Massive React community, easy to hire developers
7. **Future-Proof:** Easy to add React Native mobile app later (shared components)

### **Developer Stack:**

- **Frontend:** Next.js 14 + React 18 + TypeScript
  - **Styling:** Tailwind CSS + shadcn/ui
  - **State Management:** Zustand or React Context (simple enough)
  - **Forms:** React Hook Form + Zod validation
  - **API Calls:** Axios or native fetch with React Query
  - **Charts:** Recharts
  - **Icons:** Lucide React or Heroicons
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## **13. Additional Considerations**

### **WhatsApp-First Strategy**

- Web dashboard is secondary for advanced users
- 80% of interactions should work via WhatsApp
- Dashboard serves as:
  - Detailed analysis view
  - Historical data access
  - Complex operations (subsidy applications)



## **Offline Capability**

- Cache last 30 days of data
- Allow viewing reports offline
- Sync when connection restored

## **Localization**

- Use i18n library (next-i18next)
  - Design UI to accommodate longer text (German/Tamil can be 30% longer)
  - Cultural adaptation: Festivals, regional tax calendars
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## **14. Figma Design System Setup**

**Before coding, create in Figma:**

### **1. Design Tokens**

- Color palette
- Typography scale
- Spacing system (4px/8px grid)
- Border radius values
- Shadow styles

### **2. Component Library**

- Buttons (primary, secondary, text)
- Input fields, dropdowns
- Cards, modals

- Alert/notification styles
- Navigation components

### 3. **Responsive Breakpoints**

- Mobile: 375px - 768px
- Tablet: 768px - 1024px
- Desktop: 1024px+

### 4. **User Testing**

- Test with 5-10 MSME owners
- Focus on: Can they upload invoice in <30 seconds?
- Validate subsidy discovery flow

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## **Conclusion**

This PRD provides a complete blueprint for the Micro-CFO frontend. The Next.js stack offers the best balance of performance, developer experience, and scalability for this use case. The component-based architecture ensures you can iterate quickly while maintaining code quality.

### **Next Steps:**

1. Get stakeholder approval on this PRD
2. Create Figma mockups (use this as reference)
3. Set up Next.js project with recommended stack
4. Begin frontend development in parallel with backend AI agent work
5. Continuous user testing throughout development

Good luck building Micro-CFO! 🚀

